A Statement With Some Shape

A synthetic example

Theorem 1. *Let E be a nonempty type and $1 \le j \le k$. For every x there is a constant C such that, for every n, both $x \le C$ and $n + x \le n + C$.*

Theorem 2. *For every j there is a function x with $x(j) = j$ and $j \le x(j)$, and there is a boolean x that is true.*

Theorem 3. *Let E be nonempty. No natural number is negative, and either every n satisfies $0 \le n$ or, if $1 = 1$, some m equals 1.*

Definition 1. *A number n is a multiple of m above b if $n = mq$ for some $q \ge b$. The Fibonacci numbers are $F_0 = 0$, $F_1 = 1$ and $F_{n+2} = F_n + F_{n+1}$.*

Theorem 4. *Let s be a solution for d with bound at most 5. Then s starts at most at 5.*

Theorem 5. *If every admissible datum has a good solution, then 0 has a good solution.*

Theorem 6. *Every admissible datum d has a good solution.*

Theorem 7. *Every positive number is at least 1.*

Theorem 8. *A type with a single element has at most one element, and exactly one.*

Theorem 9. *On a probability space, the whole space has measure one.*

Theorem 10. *No solution for 0 has a negative bound.*

Theorem 11. *Two solutions s and t for the same datum start at the same point.*

Theorem 12. *If $a \le b$ and $b \le a$, then $a = b$.*

Definition 2. *A profile is a function f with $f(n) \le C$ for every n. A concrete profile is a profile that vanishes from N on and has period p below N.*

Theorem 13. *The zero profile below N is a concrete profile, and $f(n + 1) = f(n)$ for it.*

Theorem 14. *A concrete profile vanishes from N on.*

Theorem 15. *A scaled family for d and w that starts at d starts at d.*

Theorem 16. *An object with both extents has a positive first coordinate.*

Theorem 17. *If the first $n + 5$ differences of f are bounded by C on S, so are the first $n + 4$.*

Theorem 18. *A vector bounded by i is bounded by $i + 1$; vectors bounded by 7 are bounded by 8.*

Theorem 19. *The number of steps down from $k + 3$ is positive, and $\operatorname{pick}(k + 1, 0) = 1$.*

Theorem 20. *If every x with $A(x)$ has some y with $B(x, y)$, and $A(a)$ holds, then some y has $B(a, y)$.*

Theorem 21. *For every x there is y such that $x + z \le y + z$ for every z.*

Theorem 22. *If P holds, some x is 0; Q holds exactly when some n equals itself; and either some x is 1 or Q holds.*

Theorem 23. *Some map $\Phi \colon G \to H$ over $\mathbb{N}$ has both A and B.*

Theorem 24. *There is a constant C with $n \le Cn^2 + n$ for every n; and for every n there is such a constant.*

Theorem 25. *Let μ and ν be σ-finite measures and $\Phi \colon L^2(\mu \otimes \nu) \to \mathbb{R}^3$ a linear isometry. Every class F has a representative, unique almost everywhere.*

Theorem 26. *If $m \le b$ for every $m \le n$, then some real x exceeds n, or $n = 2x$ for some x.*

Theorem 27. *Let m be a σ-algebra on Ω besides the given one. If f is strongly measurable and g is measurable for m, and g is strongly measurable for the given one, then g is strongly measurable for m.*

Theorem 28. *For every n, the number that is 0 when $n = 0$ and n otherwise is at most n.*

Definition 3. *The value of the digit d_i is i, for $i = 0, \dots, 69$.*

Theorem 29. *The value of the digit d_{69} is 69.*

Theorem 30. *A digit whose constructor index is 69 has the value 69.*

Theorem 31. *Half of $k + 2$ is less than $k + 2 + k$.*

Definition 4. *Let $D_{a,b}(t, x)$ be the derivative $\partial_t^a \partial_x^b u$ at time t and point x. Then $J_x(D, k, C)$ says that $D_{0,b} \le C$ for every $b \le k$ (derivatives in space), $J_t(D, k, C)$ that $D_{a,0} \le C$ for every $a \le k$ (derivatives in time), and $J(D, k, C)$ that $D_{a,b} \le C$ for every $a + b \le k$ (all variables). The norm in space $M_x(u, t, N)$ is the largest $u(t, x)$ over $x \le N$, and the norm in time $M_t(u, x, N)$ the largest $u(s, x)$ over $s \le N$.*

Theorem 32. *If $J(D, k, C)$, then $J_x(D, k, C)$ and $J_t(D, k, C)$.*

Theorem 33. *If $M_x(u, t, N) \le C$ and $M_t(u, x, N) \le C$, then both bounds hold.*